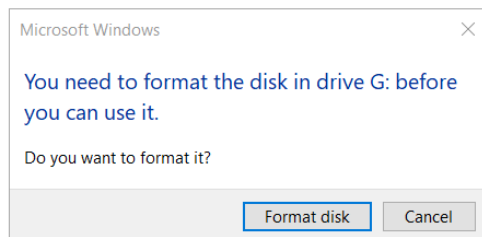


TUTORIAL NO 4: CORRUPTED HARD DISK DRIVE

You come across a damaged disk image retrieved from a suspect's laptop and you want to retrieve the content of the hard disk. The file system appears to be corrupted but hidden deep inside the broken structure lies critical information that could unlock the next steps in your investigation.



The goal of this tutorial will be to try and repair the image, knowing that the corruption comes from tampering with the OEM ID.

Data source: <https://drive.proton.me/urls/15NQK5V8B0#VAesxikOWzxP>

Question: What is an OEM ID? Of what importance is it to the file system of an OS?

Step 1: Accessing the File System

In order to access and manipulate the raw binary content of this file more efficiently on windows, you are going to be using PowerShell (using commands like “*Format-Hex C:\<partial_path>\deleted.vhd*”) or a more specialized and preferred tool like HxD (GUI hex editor). On linux or ubuntu you can analyse this image with built in command line tools like “hexdump”.



Offset (h)	00	01	02	03	04	05	06	07	08	09	0A	0B	0C	0D	0E	0F	Decoded text
00000000	33	C0	8E	D0	BC	00	7C	8E	C0	8E	D8	BE	00	7C	BF	00	SAZ0%. ZAZ0%. Z.
00000010	06	B9	00	02	FC	F3	A4	50	68	1C	06	CB	FB	B9	04	00	..üóPh..Èù^..
00000020	BD	BE	07	80	7E	00	00	7C	0B	0F	85	0E	01	83	C5	10	%%.E~..fÁ.
00000030	E2	F1	CD	18	88	56	00	55	C6	46	11	05	C6	46	10	00	ñI..^V.UEF..EF..
00000040	B4	41	BB	AA	55	CD	13	5D	72	0F	81	FB	55	AA	75	09	'A»^UI.. r..ûU^u.
00000050	F7	C1	01	00	74	03	FE	46	10	66	60	80	7E	10	00	74	Á..t.pf.f^E~..t
00000060	26	66	68	00	00	00	00	66	FF	76	08	68	00	00	68	00	afh....fÿv.h..h.
00000070	7C	68	01	00	68	10	00	B4	42	8A	56	00	8B	F4	CD	13	h..h..^B\$V.<ôí.
00000080	9F	83	C4	10	9E	EB	14	B8	01	02	BB	00	7C	8A	56	00	YfÁ.žè... .ŠV.
00000090	8A	76	01	8A	4E	02	8A	6E	03	CD	13	66	61	73	1C	FE	Šv.ŠN.Šn.İ.fas.p
000000A0	4E	11	75	0C	80	7E	00	80	0F	84	8A	00	B2	80	EB	84	N.u.E~.E..Š..eE
000000B0	55	32	E4	8A	56	00	CD	13	5D	EB	9E	81	3E	FE	7D	55	U2aŠV.İ.]èž.>p}U
000000C0	AA	75	6E	FF	76	00	E8	8D	00	75	17	FA	B0	D1	E6	64	^unÿv.è..u.û^Šad
000000D0	E8	83	00	B0	DF	E6	60	E8	7C	00	B0	FF	E6	64	E8	75	èf.^Bæ^è .°ÿadèu
000000E0	00	FB	B8	00	BB	CD	1A	66	23	C0	75	3B	66	81	FB	54	..ù..»İ.f\$Au;f.ûT
000000F0	43	50	41	75	32	81	F9	02	01	72	2C	66	68	07	BB	00	CPAu2.ù..r,fh..».
00000100	00	66	68	00	02	00	00	66	68	08	00	00	00	66	53	66	..fh....fh....fSf
00000110	53	66	55	66	68	00	00	00	00	66	68	00	7C	00	00	66	SfUfh....fh.. .f
00000120	61	68	00	00	07	CD	1A	5A	32	F6	EA	00	7C	00	00	CD	ah...İ.Z20è.. .İ
00000130	18	A0	B7	07	EB	08	A0	B6	07	EB	03	A0	B5	07	32	E4	..è..İ.è..p.2a
00000140	05	00	07	8B	F0	AC	3C	00	74	09	BB	07	00	B4	0E	CD	...<8-<t..»..'.İ
00000150	10	EB	F2	F4	EB	FD	2B	C9	E4	64	EB	00	24	02	E0	F8	..èôôÿ+Eade.\$..â
00000160	24	02	C3	49	6E	76	61	6C	69	64	20	70	61	72	74	69	\$..Invalid parti
00000170	74	69	6F	6E	20	74	61	62	6C	65	00	45	72	72	6F	72	tion table.Error
00000180	20	6C	6F	61	64	69	6E	67	20	6F	70	65	72	61	74	69	loading operati
00000190	6E	67	20	73	79	73	74	65	6D	00	4D	69	73	73	69	6E	ng system.Missin
000001A0	67	20	6F	70	65	72	61	74	69	6E	67	20	73	79	73	74	g operating syst
000001B0	65	6D	00	00	00	63	7B	9A	58	BD	FD	5C	00	00	00	02	em...C{ŠX^ÿ)\....
000001C0	03	00	07	FE	3F	0B	80	00	00	00	00	08	03	00	00	00	...p?.E.....
000001D0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
000001E0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
000001F0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	55 AA	U^

55 AA = MBR SIGNATURE

In order to access the file system, we need to understand the disk structure, so as to access the LBA which will point to the hex data of the file system. This information is contained in the MBR of the HDD.

In the MBR, the first 446 bytes is the BootCode, and the next 64 bytes is the Partition table which is of interest to us.

00000190	6E 67 20 73 79 73 74 65 6D 00 4D 69 73 73 69 6E	ng system.Missin
000001A0	67 20 6F 70 65 72 61 74 69 6E 67 20 73 79 73 74	g operating syst
000001B0	65 6D 00 00 00 63 7B 9A 58 BD FD 5C 00 00 00 02	em...c{ŠX²şý\....
000001C0	03 00 07 FE 3F 0B 80 00 00 00 00 08 03 00 00 00	...p?.€.....
000001D0	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
000001E0	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
000001F0	00 00 00 00 00 00 00 00 00 00 00 00 00 00 55 AA	U²


The Partition Table

Now knowing the structure of Partition Table, we can get the LBA address (which are the 8-11 bytes of the partition table).

So The LBA address is **80 00 00 00** , but in little endian so, it will be 0x80. Therefore, LBA address = $0x80 \times 200 = 0x10000$ (where 200 is the sector size).

Questions: What is the LBA? Why is it 80 00 00 00? What is a sector size? What is at address 0x10000?

00010000	EB 52 90 4E 54 46 00 20 20 20 20 00 02 08 00 00	ER.NTF.
00010010	00 00 00 00 00 00 F8 00 00 3F 00 FF 00 80 00 00	ø...?.ÿ.€...
00010020	00 00 00 00 00 80 00 80 00 FF 07 03 00 00 00 00	€.€.ÿ.....
00010030	55 20 00 00 00 00 00 00 02 00 00 00 00 00 00	U
00010040	F6 00 00 00 01 00 00 00 36 78 F1 B0 8F F1 B0 54	ö.....6xñ°.ñ°T
00010050	00 00 00 00 FA 33 C0 8E D0 BC 00 7C FB 68 C0 07	ú3ÀŽĐ¼. ôhÀ.
00010060	1F 1E 68 66 00 CB 88 16 0E 00 66 81 3E 03 00 4E	..hf.Ë^...f.>..N
00010070	54 46 53 75 15 B4 41 BB AA 55 CD 13 72 0C 81 FB	TFSu. 'A»²UÍ.r..û
00010080	55 AA 75 06 F7 C1 01 00 75 03 E9 DD 00 1E 83 EC	U²u.÷Á..u.éÝ..fì
00010090	18 68 1A 00 B4 48 8A 16 0E 00 8B F4 16 1F CD 13	.h.. 'HŠ...<ô..Í.
000100A0	9F 83 C4 18 9E 58 1F 72 E1 3B 06 0B 00 75 DB A3	ŸfÄ.žX.rá;...uÔ£
000100B0	0F 00 C1 2E 0F 00 04 1E 5A 33 DB B9 00 20 2B C8	..Á.....Z3Û². +È
000100C0	66 FF 06 11 00 03 16 0F 00 8E C2 FF 06 16 00 E8	fÿ.....ŽÂÿ...è
000100D0	4B 00 2B C8 77 EF B8 00 BB CD 1A 66 23 C0 75 2D	K.+Èwì,.»Í.f#Àu-
000100E0	66 81 FB 54 43 50 41 75 24 81 F9 02 01 72 1E 16	f.ûTCPAu\$.ù..r..
000100F0	68 07 BB 16 68 52 11 16 68 09 00 66 53 66 53 66	h.».hR..h..fSfSf
00010100	55 16 16 16 68 B8 01 66 61 0E 07 CD 1A 33 C0 BF	U...h..fa..Í.3À;

We are now in the NTFS file system.

Step 02: The OEM ID which consists of a string of characters (often 8 bytes) that used by the operating system and other software to understand how to interact with the hardware. So go with the structure of the file system to know what is the corrupted OEM ID, you can notice it as it is NTF without S, but it is better to know what the location exactly.

```
000 EB 52 90 4E 54 46 53 20 20 20 20 00 02 08 00 00
010 00 00 00 00 00 F8 00 00 3F 00 FF 00 00 08 00 00
020 00 00 00 00 80 00 80 00 FF EF 7F 07 00 00 00 00

EB 52 90 4E 54 46 00 20 20 20 20 00 02 08 00 00 ER.NTF. ....
```

MODIFY TO XX

NTF ; NO 'S'

Challenge: What should be the value XX to repair this OEM ID?

Modify the byte XX accordingly so as to repair the broken file system

Step 03: After modifying the byte, try remounting the .vhd image and see if it works.

Report: In your own words and understanding, write up a report (no more than 2 pages), explaining your results and what you have learned from this tutorial.